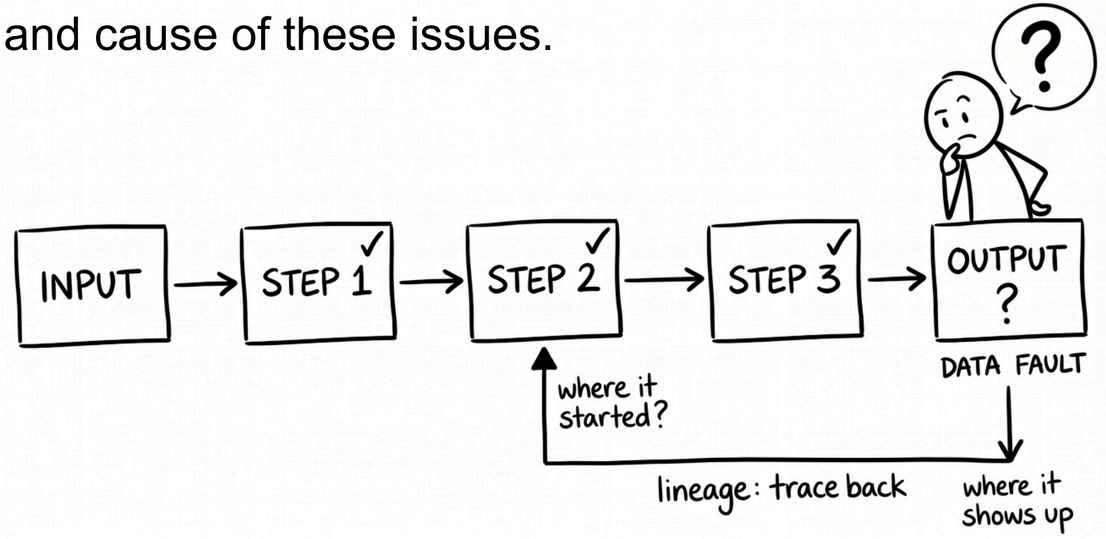
Lineage-Aware Root-Cause Analysis for Data Engineering Pipelines

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Summary: **Data pipelines often fail** silently, this means that every step reports success, but the data is wrong. Spotting that something is wrong is well supported but finding **where** it started, and **why**, is not.
This work tests whether giving a language model the **pipeline's lineage** helps it find the origin and cause of these issues.

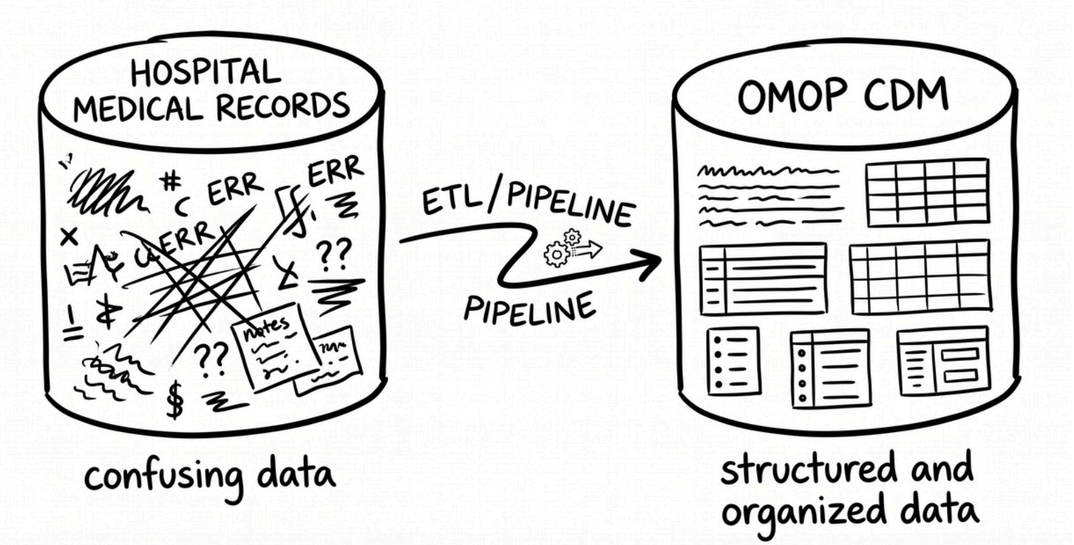


Objectives / Work done / Results

- The objective of this work is to test **how much lineage influences** the ability to find the root cause of a data fault.
- To test this we will need **fault injection**, so the true cause is always known and accuracy can be measured.

- This dissertation will focus on **OMOP CDM** data pipelines. It's a real ETL use case with several steps, where data quality matters.

- To work with this type of data we already asked and received access to a Medical Database (MIMIC).



Next Steps / Challenges / Bottlenecks

- Moving forward the objective is to continue the **State of the Art.**



- A possible **challenge** is defining "the root cause" when a fault crosses several steps, or when more than one fault is involved.